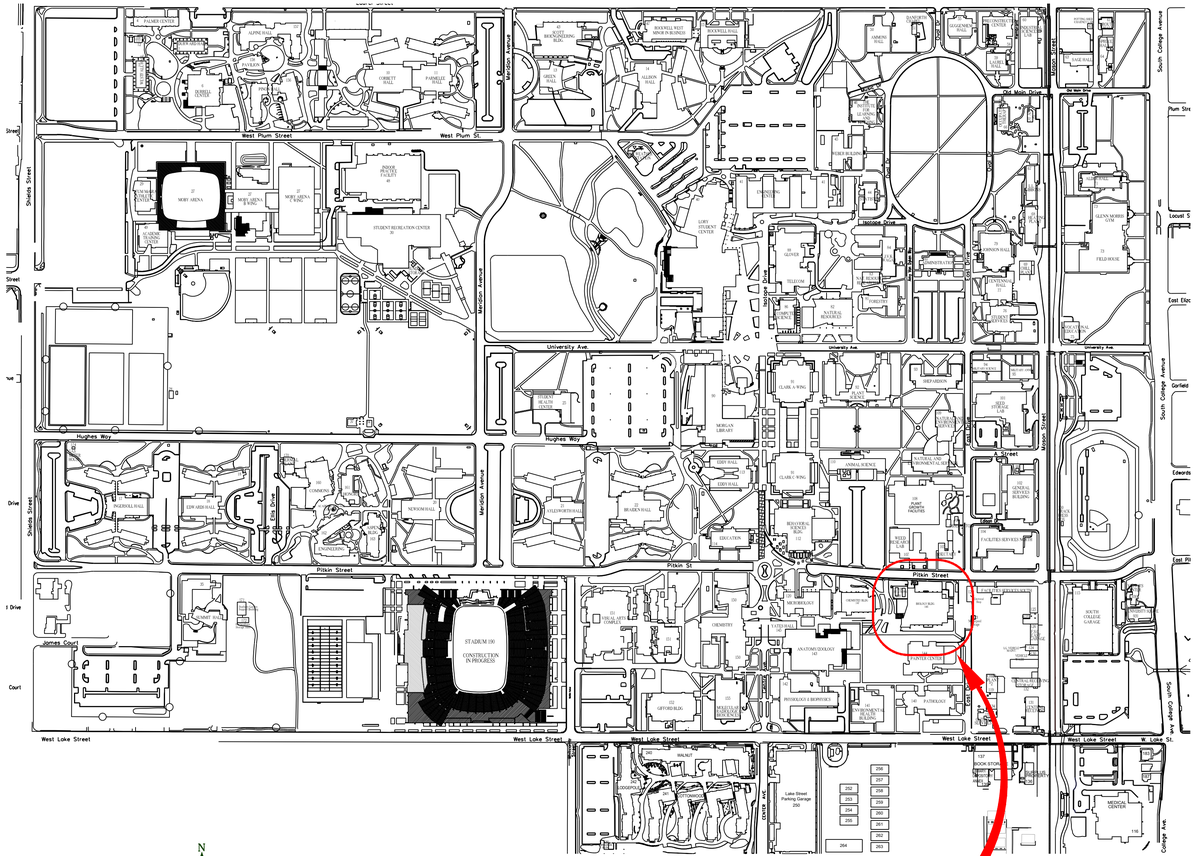
Colorado State University

BIOLOGY
BUILDING NUMBER: 146
PROJECT NUMBER: 260316D



N
W E
S

LOCATION MAP
NOT TO SCALE

PROJECT LOCATION

SCOPE

REPLACE DAMAGED EXTERIOR DUCT INSULATION ON ALL ROOF MOUNTED EXHAUST DUCT WORK TO HRU'S. REPAIR/REPLACE STAINLESS STEEL EXHAUST DUCT STACK ON EF-3.

DRAWING INDEX

G0.0	TITLE SHEET
M0.0	MECHANICAL LEGEND, NOTES AND SPECIFICATIONS
M1.0	MECHANICAL PLAN

CODES

2024 IBC, INTERNATIONAL BUILDING CODE
2024 IEBC, INTERNATIONAL EXISTING BUILDING CODE
2024 IMC, INTERNATIONAL MECHANICAL CODE
2021 IPC, INTERNATIONAL PLUMBING CODE
2026 NEC, NATIONAL ELECTRICAL CODE
2024 IFC, INTERNATIONAL FIRE CODE
2021 IFGC, INTERNATIONAL FUEL GAS CODE
ICC A117.1-2017, ACCESSIBLE AND USABLE BUILDINGS AND FACILITIES
2024 IECC, INTERNATIONAL ENERGY CONSERVATION CODE

COLORADO STATE CONSTRUCTION STANDARDS:

www.fm.colostate.edu/constr_standards

CONTRACTOR SHALL PURCHASE PARKING PERMITS FROM PARKING AND TRANSPORTATION SERVICES (970-491-7041, 1508 CENTER AVE.) FOR ALL VEHICLES DURING CONSTRUCTION. PROVIDE ADEQUATE SIGNAGE THAT READS "CONSTRUCTION PARKING" AT ALL PARKING SPACES RENTED DURING CONSTRUCTION.

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Sheet Information

Sheet Title:

TITLE SHEET

Sheet Number:

G0.0

DUCTWORK LEGEND

SYMBOL	DESCRIPTION
	90° ELBOW DOWN (ROUND DUCT ONLY)
	ROUND 90° ELBOW UP (ROUND DUCT ONLY)
	ROUND RADIUS ELBOW
	90° STRAIGHT TEE
	45° BRANCH
	ROUND BRANCH FROM RECTANGULAR: WITH DAMPER / WITHOUT DAMPER
	SIZE OR SHAPE TRANSITION
	RECTANGULAR TO ROUND TRANSITION
	ROUND FLEXIBLE DUCT
	90° ELBOW DN (NEGATIVE PRESSURE)
	90° ELBOW DN (POSITIVE PRESSURE)
	90° ELBOW UP (NEGATIVE PRESSURE)
	90° ELBOW UP (POSITIVE PRESSURE)
	90° RADIUS ELBOW (RECTANGULAR DUCT)
	90° RADIUS ELBOW W/TURNING VANES
	SQUARE DUCT SPLIT
	ROUND DUCT SPLIT
	SPLIT BRANCH TAKE-OFF WITH SQUARE ELBOW & SPLITTER DAMPER
	SPLIT BRANCH TAKE-OFF WITH RADIUS ELBOW & SPLITTER DAMPER
	DUCT CAP
	POSITIVE PRESSURE RISER, TYPICALLY SUPPLY.
	NEGATIVE PRESSURE RISER, TYPICALLY RETURN, EXHAUST OR OUTSIDE AIR.
	SUPPLY DIFFUSER
	RETURN / EXHAUST GRILLE
r/s	COMBINATION FIRE AND SMOKE DAMPER
	FIRE DAMPER
	SMOKE DAMPER
le	MOTOR OPERATED DAMPER (MOD)
h	MANUAL VOLUME DAMPER, SINGLE BLADE DAMPER (SBD) FOR ROUND OR <10" TALL, OPPOSED BLADE DAMPER (OBD) >10" TALL.
BDD	BACKDRAFT DAMPER
SD	SMOKE DETECTOR
24×36	DUCT SIZE: FIRST NUMBER IS PLAN WIDTH, SECOND NUMBER IS DEPTH.

PLAN SYMBOLS

	CONTROL / ELECTRIC PANEL
CO2	CARBON DIOXIDE SENSOR
CO	CARBON MONOXIDE SENSOR
T	THERMOSTAT
TS	REMOTE TEMPERATURE SENSOR
H	HUMIDISTAT
SP	DUCT STATIC PRESSURE SENSOR
P	ROOM PRESSURE SENSOR
EPO	EMERGENCY POWER OFF SWITCH
TC	TEMPERATURE CONTROLLER
	SECTION CUT LETTER/ SHEET SHOWN ON.
	POINT OF DISCONNECTION
	POINT OF NEW CONNECTION
	ACCESS PANEL
(E)	EXISTING ITEM
(N)	NEW ITEM
	EXISTING WORK, LIGHT LINEWEIGHT
	NEW WORK, DARKER LINEWEIGHT
	ITEM TO BE REMOVED (HATCHED)

NOTES

1. ALL SYMBOLS, ABBREVIATIONS, AND DESIGNATIONS ON LEGEND SHEET ARE NOT NECESSARILY USED ON THIS PROJECT.
2. THIS DRAWING SET MAY CONSIST OF DATA GENERATED, IN PART, BY OTHER PARTIES. NOT ALL SYMBOLOGIES AND NOTATION CONVENTIONS OCCURRING IN THIS DRAWING SET ARE NECESSARILY DEFINED ON THESE LEGENDS. CONSULT THE ENGINEER IN THE EVENT SYMBOLOGY OR NOTATION INTERPRETATION IS REQUIRED.

GENERAL MECHANICAL NOTES

1. ALL WORK SHALL COMPLY WITH APPLICABLE STATE AND LOCAL CODES, LAWS, INDUSTRY STANDARDS AND CSU'S BUILDING CONSTRUCTION STANDARDS AND TECHNICAL STANDARDS. COORDINATE THE MECHANICAL WORK WITH THE CSU PM AND/OR GENERAL CONTRACTOR AS APPLICABLE, AS WELL AS ALL OTHER TRADES ON THE PROJECT.
2. FURNISH ALL LABOR, MATERIALS AND EQUIPMENT, AND PERFORM ALL OPERATIONS NECESSARY FOR THE INSTALLATION OF COMPLETE AND OPERATING MECHANICAL SYSTEMS INDICATED ON THESE DRAWINGS. ALL WORKMANSHIP AND MATERIALS SHALL BE TO CSU STANDARDS. CODE INSPECTIONS WILL BE CONDUCTED PER THE REQUIREMENTS OF THE LOCAL CODE AUTHORITY.
3. ALL CORRIDORS AND OTHER PROJECT ACCESS AREAS SHALL BE KEPT CLEAN AT ALL TIMES. DUST BARRIERS, EXTENDING FROM FLOOR THROUGH ANY DROPPED CEILINGS UP TO STRUCTURE ABOVE, SHALL BE USED TO ENCLOSE THE WORK AREA WHEN REMOVING AND CUTTING MASONRY OR WHEN DOING OTHER WORK THAT CREATES DUST AND DEBRIS. ALL TRASH AND DEBRIS ARE TO BE REMOVED FROM THE SITE AND DISPOSED OF PROPERLY. DO NOT USE CSU DUMPSTERS. PROTECT ALL INTERIOR SURFACES AND FIXTURES FROM CONSTRUCTION DAMAGE AND RETURN TO ORIGINAL CONDITION.
4. MECHANICAL DRAWINGS ARE DIAGRAMMATIC, AND NOT INTENDED TO BE SCALED FOR EXACT MEASUREMENTS, BUT SHOULD BE FOLLOWED AS CLOSELY AS ACTUAL CONSTRUCTION OF THE BUILDING WILL PERMIT. WHERE CONDITIONS REQUIRE REASONABLE CHANGES TO THE MECHANICAL WORK, MAKE SUCH CHANGES WITHOUT ADDITIONAL COST TO THE OWNER. MAKE NO MAJOR CHANGE NOR DECREASE SIZE OF ANY DUCTWORK, PIPING OR EQUIPMENT WITHOUT WRITTEN PERMISSION. DEVIATION FROM THE PLANS AND SPECIFICATIONS MADE WITHOUT CONSENT OF THE ENGINEER SHALL RELIEVE THE ENGINEER OF RESPONSIBILITY FOR ALL CONSEQUENCES ARRIVING OUT OF SUCH CHANGES.
5. EXISTING SYSTEMS AND CONDITIONS SHOWN ARE BASED ON ORIGINAL OR REMODEL DRAWINGS AND/OR LIMITED SITE INVESTIGATION, AND HAVE NOT BEEN FULLY VERIFIED FOR ACCURACY. FIELD VERIFY ALL EXISTING CONDITIONS AND DIMENSIONS. NOTIFY CSU PM AND ENGINEER IF ANY CONDITIONS PREVENT WORK FROM PROCEEDING AS GENERALLY INDICATED ON THESE DRAWINGS. VERIFY PROPER OPERATION OF ANY AND ALL EXISTING EQUIPMENT NOTED TO REMAIN. NOTIFY CSU PM AND ENGINEER OF ANY DEFICIENCIES THAT PREVENT EQUIPMENT FROM OPERATING AS INTENDED.
6. CUTTING OF STRUCTURAL MEMBERS IS NOT ALLOWED.
7. WHERE MOUNTING HEIGHTS ARE NOT DETAILED OR DIMENSIONED, INSTALL MECHANICAL SERVICES AND OVERHEAD EQUIPMENT TO PROVIDE THE MAXIMUM HEADROOM POSSIBLE, AND IN ACCORDANCE WITH MINIMUM REQUIRED CLEARANCES.
8. CONTRACTOR SHALL PROVIDE PRODUCT SUBMITTAL DATA FOR REVIEW. PROVIDE INFORMATION ON ALL EQUIPMENT AS SCHEDULED ON THE DRAWINGS. REFERENCE EACH PIECE OF EQUIPMENT AS LABELED ON EQUIPMENT SCHEDULES, AND INCLUDE PERFORMANCE DATA, ACCESSORIES AND CONTROLS DIAGRAMS AS NECESSARY. QUANTITIES WILL NOT BE CHECKED OR VERIFIED. IT IS THE CONTRACTOR'S RESPONSIBILITY TO PROVIDE PROPER QUANTITIES REQUIRED TO COMPLETE THE JOB. ENGINEER'S ACCEPTANCE OF SUBMITTALS WILL NOT RELIEVE THE CONTRACTOR FROM HIS RESPONSIBILITY FOR ANY ERRORS OR OMISSIONS IN SUBMITTALS.
9. EQUIPMENT AND MATERIALS LISTED ON THE DRAWINGS AND DETAILED IN EQUIPMENT SCHEDULES ARE TO BE CONSIDERED THE BASIS OF DESIGN. WHERE ALTERNATE MANUFACTURERS ARE LISTED, THE CHOICE OF THESE IS OPTIONAL WITH THE CONTRACTOR. EQUIPMENT AND MATERIALS NOT LISTED AS EQUIVALENTS MAY BE PROPOSED AS DEDUCTIVE ALTERNATES TO SPECIFIED ITEMS, AND MUST BE SUBMITTED WITH FULL DESCRIPTIVE DATA, COST DEDUCTION AND ANY DEVIATIONS FROM SPECIFIED ITEMS. CONTRACTOR IS RESPONSIBLE FOR ANY CHANGES AND COSTS TO ACCOMMODATE SUBSTITUTED MATERIALS OR EQUIPMENT.
10. DURING THE PROCESS OF THE WORK, MAINTAIN A COMPLETE AND ACCURATE RECORD OF THE ACTUAL INSTALLATION, AND LOCATIONS OF THE MECHANICAL SYSTEMS. TRANSFER ALL RECORD DATA TO PRINTS OF THE ORIGINAL DRAWINGS, AND FORWARD TO CSU PM UPON COMPLETION OF THE PROJECT.
11. GUARANTEE ALL WORKMANSHIP, MATERIALS, EQUIPMENT AND PROPER OPERATION FOR A PERIOD OF ONE YEAR FROM THE DATE OF FINAL ACCEPTANCE FROM THE OWNER. EACH WARRANTY FOR LONGER THAN ONE YEAR (THAT COMES WITH EQUIPMENT USED ON THE JOB) SHALL BE PASSED ON TO THE OWNER WITH DATES OF START AND END OF THE WARRANTY.
12. UPON FINAL COMPLETION OF WORK, ALL FIXTURES AND EQUIPMENT SHALL BE THOROUGHLY CLEANED AND MADE READY FOR UTILIZATION BY THE OWNER. ALL FILTERS SHALL BE REPLACED WITH NEW PRIOR TO OWNER TURNOVER.

23 07 00 - HVAC INSULATION

PART 1 - GENERAL

1.1 QUALITY ASSURANCE:

A. MANUFACTURER'S QUALIFICATIONS: FIRMS REGULARLY ENGAGED IN MANUFACTURE OF MECHANICAL INSULATION PRODUCTS AND SYSTEMS, OF TYPES AND SIZES REQUIRED, WHOSE PRODUCTS HAVE BEEN IN SATISFACTORY USE IN SIMILAR SERVICE FOR NOT LESS THAN 3 YEARS.

B. INSTALLER'S QUALIFICATIONS: FIRM WITH AT LEAST 5 YEARS SUCCESSFUL INSTALLATION EXPERIENCE ON PROJECTS WITH MECHANICAL INSULATIONS SIMILAR TO THAT REQUIRED FOR THIS PROJECT.

C. FLAME/SMOKE RATINGS: PROVIDE COMPOSITE MECHANICAL INSULATION (INSULATION, JACKETS, COVERINGS, SEALERS, MASTICS AND ADHESIVES) WITH FLAME-SPREAD INDEX OF 25 OR LESS, AND SMOKE-DEVELOPED INDEX OF 450 OR LESS, AS TESTED BY ASTM E 84.

1.2 SUBMITTALS:

A. PRODUCT DATA: SUBMIT MANUFACTURER'S TECHNICAL PRODUCT DATA AND INSTALLATION INSTRUCTIONS FOR EACH TYPE OF MECHANICAL INSULATION. SUBMIT SCHEDULE SHOWING MANUFACTURER'S PRODUCT NUMBER, K-VALUE, THICKNESS, DENSITY, AND FURNISHED ACCESSORIES FOR EACH MECHANICAL SYSTEM REQUIRING INSULATION. SUBMIT DETAIL PRODUCT INFORMATION AND INSTALLATION INFORMATION FOR ALL JACKETING SYSTEMS SPECIFIED IN THIS SECTION.

1.3 DELIVERY, STORAGE, AND HANDLING:

A. DELIVER INSULATION, COVERINGS, CEMENTS, ADHESIVES, AND COATINGS TO SITE IN ORIGINAL PACKAGING UNBROKEN AND LABELED WITH MANUFACTURER'S NAME, PRODUCT BRAND NAME AND TYPE.

B. STORE MATERIALS IN CLEAN, DRY AREA IN MANUFACTURER'S UNOPENED PACKAGING UNTIL READY FOR INSTALLATION AND IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS AND TEMPERATURE RECOMMENDATIONS.

C. PROTECT INSULATION AGAINST DIRT, WATER, AND CHEMICAL AND MECHANICAL DAMAGE. DO NOT INSTALL DAMAGED OR WET INSULATION; REMOVE FROM PROJECT SITE.

PART 2 - PRODUCTS

2.1 MANUFACTURERS:

A. MANUFACTURER- SUBJECT TO COMPLIANCE WITH REQUIREMENTS, PROVIDE PRODUCT BY ONE OF THE FOLLOWING:

1. MECHANICAL INSULATION:
   a. JOHNS MANVILLE CORP.
   b. OWENS-CORNING FIBERGLAS CORP.
   c. KNAUF FIBER GLASS
   d. MANSON
   e. CERTAINTEED
2. JACKETING & COVERING PRODUCTS:
   a. POLYGUARD, ALUMAGUARD

2.2 DUCTWORK INSULATION MATERIALS:

A. FOIL-FACED POLYISOCYANURATE RIGID FOAM BOARD: ASTM C1289, TYPE 1, CLASS 1, 2" THICKNESS, MINIMUM R VALUE 12. SERVICE TEMPERATURE OF -100 TO 250 DEGREES F, WATER VAPOR PERMEABILITY RATING OF 0.05 PERM INCHES OR LESS. JOHNS MANVILLE XSPECT ISOFORM APF BOARD OR EQUAL.

B. JACKET FOR DUCTWORK INSULATION: LAMINATED JACKETING SYSTEM.

1. THE JACKETING SHALL BE A SELF-HEALING, PUNCTURE RESISTANT COMPOSITE MEMBRANE ROLL CONSISTING OF A MULTI-PLY EMBOSSED UV-RESISTANT ALUMINUM FOIL/POLYMER LAMINATE TO WHICH IS APPLIED A LAYER OF RUBBERIZED ASPHALT. MEMBRANES ARE "PEEL AND STICK" AND SUITABLE FOR USE ON INSULATED DUCT. 54-MIL THICKNESS, PERMEANCE OF 0.00. ALUMAGUARD OR EQUAL.

C. DUCTWORK INSULATION ACCESSORIES: PROVIDE MECHANICAL FASTENERS, BANDS, WIRES, TAPE, ANCHORS, CORNER ANGLES AND SIMILAR ACCESSORIES AS RECOMMENDED BY INSULATION MANUFACTURER FOR APPLICATIONS INDICATED.

D. DUCTWORK INSULATION COMPOUNDS: PROVIDE CEMENTS, ADHESIVES, COATINGS, SEALERS, PROTECTIVE FINISHES AND SIMILAR COMPOUNDS AS RECOMMENDED BY INSULATION MANUFACTURER FOR APPLICATIONS INDICATED.

3.1 DUCTWORK SYSTEM INSULATION:

A. APPLICATION REQUIREMENTS: INSULATE THE FOLLOWING DUCTWORK:

1. ALL EXTERIOR DUCT BETWEEN ROOF PENETRATION AND ENERGY RECOVERY UNITS. SEE ATTACHED DRAWING.

3.2 INSTALLATION OF DUCTWORK INSULATION AND JACKETING:

A. GENERAL: INSTALL INSULATION AND JACKETING PRODUCTS IN ACCORDANCE WITH MANUFACTURER'S WRITTEN INSTRUCTIONS, AND IN ACCORDANCE WITH RECOGNIZED INDUSTRY PRACTICES TO ENSURE THAT INSULATION IS SECURELY ATTACHED TO DUCT, AND SERVES ITS INTENDED PURPOSE.

B. INSTALL INSULATION MATERIALS WITH SMOOTH AND EVEN SURFACES.

C. CLEAN AND DRY DUCTWORK PRIOR TO ATTACHING INSULATION. EXPOSED FOIL FACING OF INSULATION MUST BE CLEAN, DRY AND FREE OF OIL FILMS BEFORE ADHERING JACKETING SYSTEM.

D. USE MECHANICAL FASTENERS OR ADHESIVE TO SECURE INSULATION TO DUCT PER MANUFACTURER'S INSTRUCTIONS. SPACE FASTENERS PER INSULATION MANUFACTURERS INSTRUCTIONS. BUTT INSULATION JOINTS FIRMLY TOGETHER TO ENSURE COMPLETE AND TIGHT FIT OVER SURFACES TO BE COVERED.

E. FOLLOW INSULATION MANUFACTURER'S INSTRUCTION FOR INSULATING OVER DUCT JOINTS AND FLANGES.

F. TOP SURFACE OF DUCT INSULATION MUST BE SLOPED TO ALLOW MOISTURE RUNOFF AND PREVENT POOLING ON TOP OF DUCT. PROVIDE TAPERED INSULATION BOARDS WITH SHORT SIDE OF BOARD AT MINIMUM THICKNESS TO MEET CODE REQUIRED R-VALUE OF INSULATION.

G. FOLLOW ALL JACKETING MANUFACTURER'S INSTALLATION INSTRUCTIONS. ADHERE TO MANUFACTURER'S TEMPERATURE RECOMMENDATIONS FOR INSTALLATION. NOTIFY CSU PM IF OUTDOOR TEMPERATURES WILL DELAY OR PREVENT INSULATION WORK FROM PROCEEDING.

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10/05/2026
PROFESSIONAL ENGINEER

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DUCT INSULATION DAMAGE

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Sheet Title:

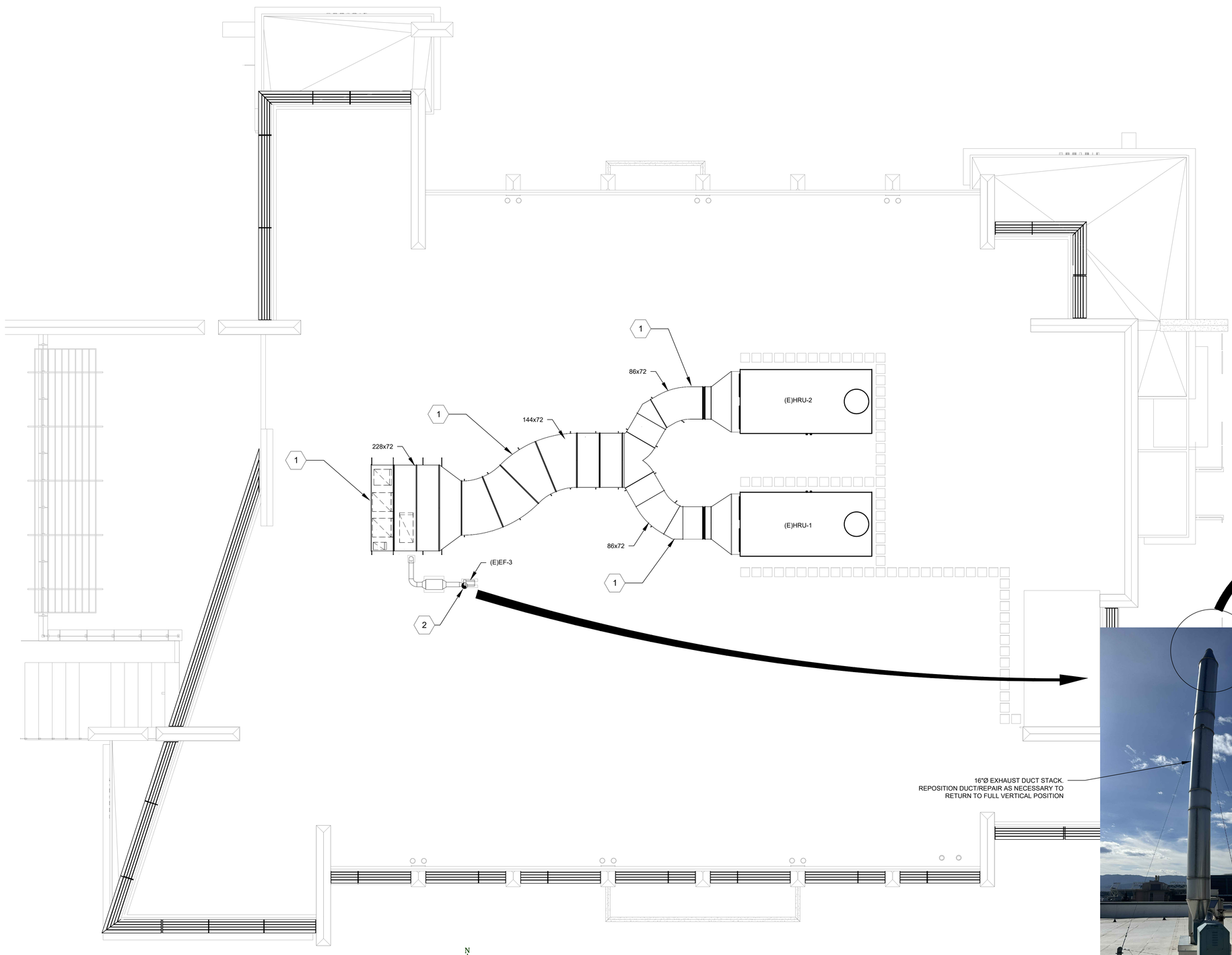
MECHANICAL LEGEND, NOTES AND SPECIFICATIONS

Sheet Number:

M0.0

KEYNOTES:

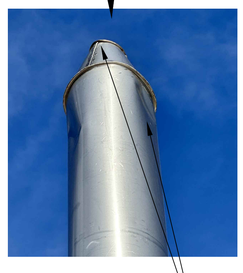
1. REMOVE AND REPLACE ALL EXTERIOR DUCT INSULATION ON ENTIRETY OF ROOF MOUNTED DUCTS BETWEEN HRU'S AND ROOF PENETRATION. SEE SPECIFICATIONS ON SHEET M0.0 FOR INSULATION AND PROTECTIVE JACKETING REQUIREMENTS.
2. EXISTING EXHAUST STACK FROM EF-3 TO BE REPAIRED AS NECESSARY. SEE PICS ON THIS SHEET. RETURN DUCT TO FULL VERTICAL POSITION, AND REPAIR/REPLACE TOP OF STACK AS NECESSARY WITH MATCHING STAINLESS STEEL DUCT.



ROOF MECHANICAL PLAN
SCALE: 3/32" = 1'-0"

16"Ø EXHAUST DUCT STACK.
REPOSITION DUCT/REPAIR AS NECESSARY TO
RETURN TO FULL VERTICAL POSITION





REPLACE TOP OF DAMAGED
DUCT STACK WITH MATCHING
DIAMETER STAINLESS
STEEL DUCT. INSPECT TOP
EXIT CONE, AND REPLACE WITH
NEW IF DAMAGED. EXIT CONE IS 7"Ø

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MECHANICAL PLAN

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